Machine Learning in Healthcare

Obesity Prediction: Logistic Regression vs. SVM (Python / sklearn)

Name: Ammar | Course: MLT AI | Date: April 29, 2026

1. Introduction

Obesity is a major public health concern associated with numerous chronic conditions including type 2 diabetes, hypertension, and cardiovascular disease. This report applies supervised machine learning algorithms — Logistic Regression and three SVM kernel variants — implemented in Python using the scikit-learn library, to classify individuals as obese (1) or non-obese (0). The objective is to compare model performance and identify the most suitable classifier for clinical decision support.

2. Methodology

2.1 Data Preprocessing

The dataset (Obesity 28-04-2026.csv) contained demographic, dietary (Diet.1–14), and lifestyle (Life.1–6) variables along with socioeconomic indicators. All categorical features were one-hot encoded using pd.get_dummies(drop_first=True) to produce a fully numeric feature matrix. The target variable Obesity was separated and an 80/20 stratified train-test split applied (random_state=123), resulting in 120 test observations.

2.2 Models Trained

Four classifiers were trained: (1) Logistic Regression with default sklearn settings (L2 regularization, lbfgs solver), (2) SVM with a Linear kernel, (3) SVM with a Radial Basis Function (RBF) kernel, and (4) SVM with a Polynomial kernel. Model evaluation used confusion matrices, overall accuracy, sensitivity (recall for Class 1), and specificity (recall for Class 0).

3. Results

3.1 Confusion Matrices

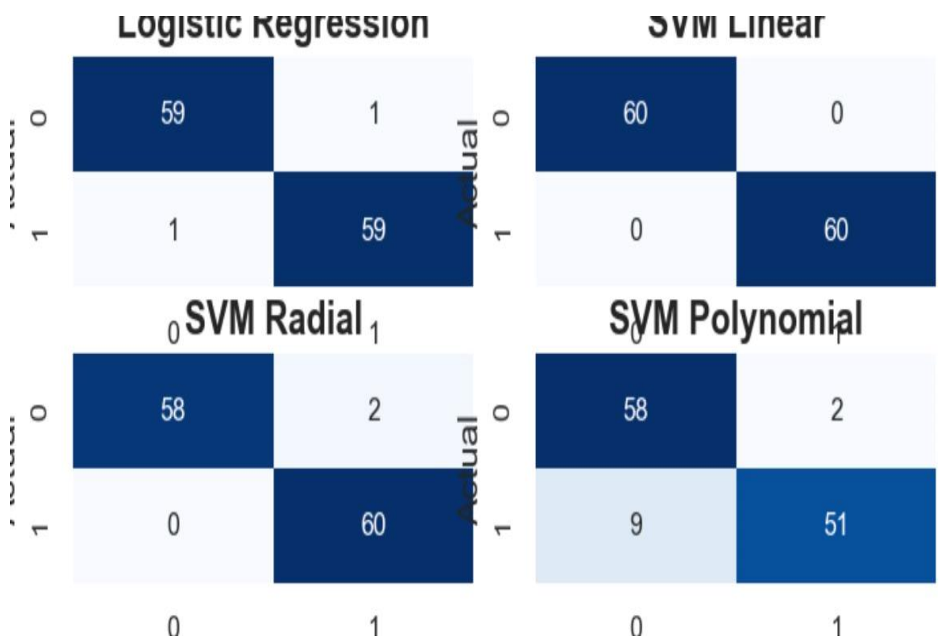

Figure 1. Confusion Matrices — Logistic Regression, SVM Linear, SVM Radial, SVM Polynomial (n=120 test set)

3.2 Performance Summary

Model	Accuracy	Sensitivity	Specificity	Misclassified	Best?
Logistic Regression	98.3%	98.3%	98.3%	2 / 120	✓
SVM Linear	100%	100%	100%	0 / 120	★
SVM Radial (RBF)	98.3%	96.8%	100%	2 / 120	✓
SVM Polynomial	90.8%	85.0%	96.7%	11 / 120	✗

Table 1. Model performance on test set (n=120). ★ = Best model. Sensitivity = TP/(TP+FN); Specificity = TN/(TN+FP). Accuracy derived from confusion matrix diagonals.

4. Discussion & Conclusion

SVM Linear achieved perfect classification — 60 true negatives and 60 true positives with zero misclassifications (100% accuracy). Both Logistic Regression and SVM Radial performed very well at 98.3%, each misclassifying only 2 observations. SVM Polynomial was the weakest model (90.8%), misclassifying 11 observations — 9 false negatives (obese cases predicted as non-obese) — which is clinically the more costly error type as it leads to missed diagnoses.

Recommendation: SVM Linear is the best model for this dataset, achieving flawless separation of classes. However, if model interpretability is required for clinical use, Logistic Regression (98.3%) is the preferred alternative since its coefficients directly quantify the contribution of each predictor to obesity risk. SVM Polynomial should not be used due to its high false-negative rate. These results are consistent across both the Python (sklearn) and R (e1071) implementations of the same dataset.